

# The phone on the head

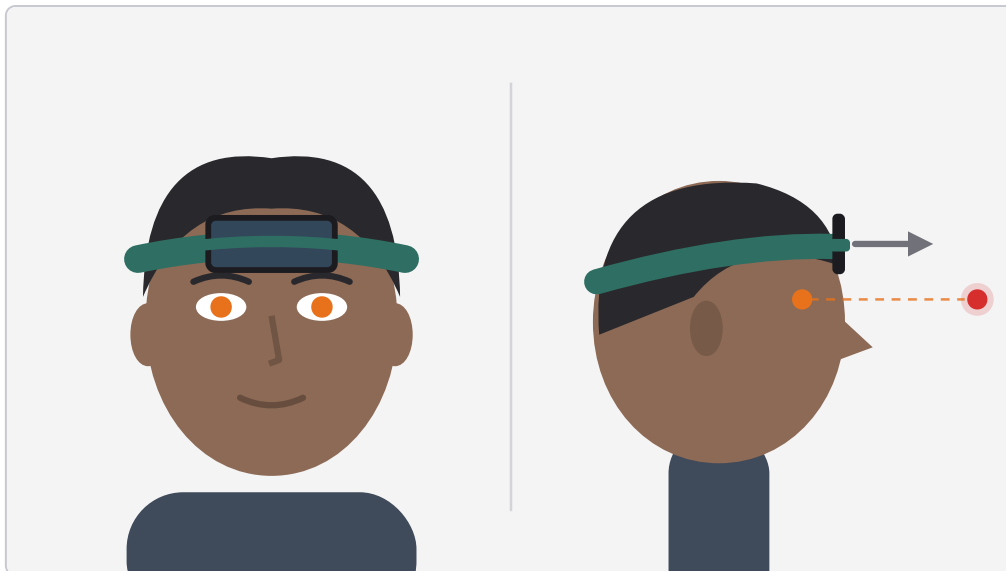
## A quick trainHIT guide to examining all six canals with a phone

With the phone strapped to a classmate's head, the impulses are real: the weight of the head, the neck that brakes and the hands that slip. trainHIT reads the turn from the phone's gyroscope, draws the face on the computer and places the eye with its model, healthy or with the pathology chosen in the **Simulator**. This way the **vertical canals** can be practised too, which the webcam cannot measure.

**This is training, not an exam.** The face and the eye are simulated: what is measured is the examiner's technique. Not for use with patients. Small impulses (10 to 20°), and never on someone with neck pain or a neck injury.

### 1. Where the phone goes

**On the forehead, in landscape, with the screen facing forward** and the back of the phone against the skin, above the eyebrows. That way the app works out by itself which way the nose points.

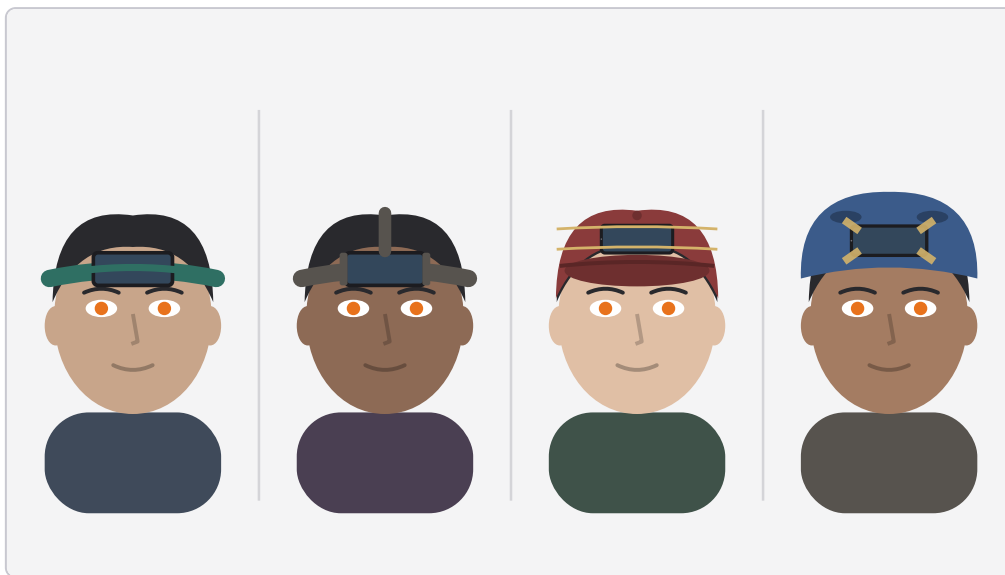


*Front and side views: the phone in landscape on the forehead, screen facing forward, held by an elastic band*

What matters:

- **It must not wobble.** If the phone moves on the skin, it adds turns the head never made and the impulse comes out with a rebound or off the plane. A piece of non-slip kitchen mat (the rubber mesh) between forehead and phone helps a lot.
- **It must not cover the eyes** or the eyebrows: above them.
- **Firm but not tight.** A phone weighs about 200 g: a well-placed band is enough.
- **Screen on.** trainHIT asks the phone not to sleep while the link lasts; a charged battery helps.

## 2. Home-made ways to hold it



*Four ways: elastic band, head-torch harness, cap with rubber bands and helmet with tape*

| With                                                      | How                                                                                | Watch out                                   |
|-----------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Wide elastic band</b> (sports or hair band, 4 to 6 cm) | The phone between the band and the forehead; if long, a second turn over it.       | Narrow bands let it rotate: wide is better. |
| <b>Head-torch harness</b>                                 | Remove the torch; the holder grips the phone. The top strap stops it sliding down. | The steadiest option.                       |
| <b>Sewing elastic</b> (60 cm × 3 cm, sewn into a ring)    | Adjust it to the head; a fabric pocket for the phone can be sewn on.               | Cheap and washable: lasts a whole course.   |
| <b>Tight cap + two rubber bands</b>                       | The phone on the front of the cap, with two rubber (or hair) bands around it.      | A loose cap slips: tighten it at the back.  |
| <b>Bike helmet + velcro or tape</b>                       | The phone stuck to the front of the helmet, strap well fastened.                   | Tape on the case, not on the phone.         |
| <b>Elastic bandage</b> (first-aid kit)                    | Two or three turns around the head over the phone.                                 | Not tight: a head is not an ankle.          |

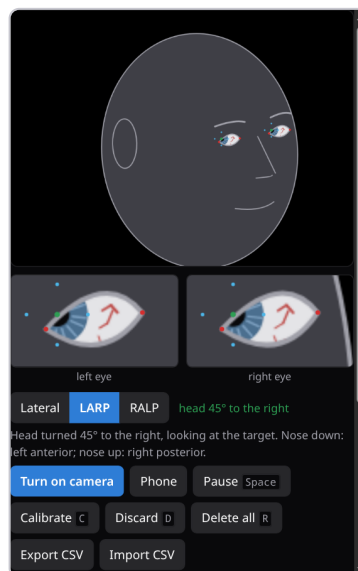
**The test before starting:** with the head still, push the phone sideways with a finger. If it moves on the skin, adjust or change the mount.

### 3. Link and center

1. On the computer, press **Phone** and **Show the QR**.
2. Scan the QR with the phone and tap **Use this phone as the head** *before* strapping it on: the sensor permission needs a tap.
3. Strap the phone to the forehead.
4. With the head straight and looking at the target (the computer's camera or a spot on the wall), press **Center** on the computer. That is "facing forward".

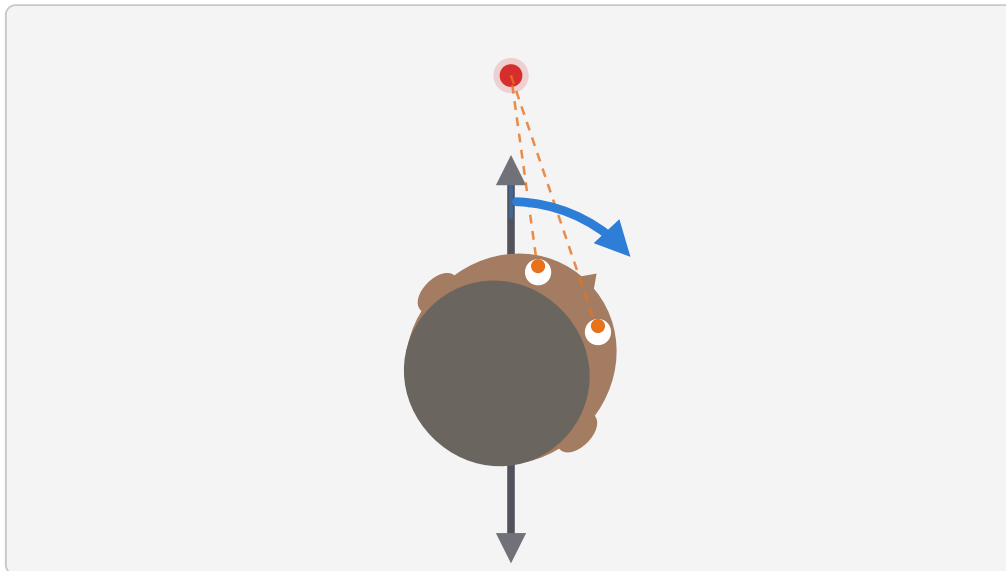
### 4. Examining each plane

With the phone linked, the computer shows the selector **Lateral · LARP · RALP**. Below it says how to place the head, and next to it how far the head is turned right now: **green** when it is where the plane asks.



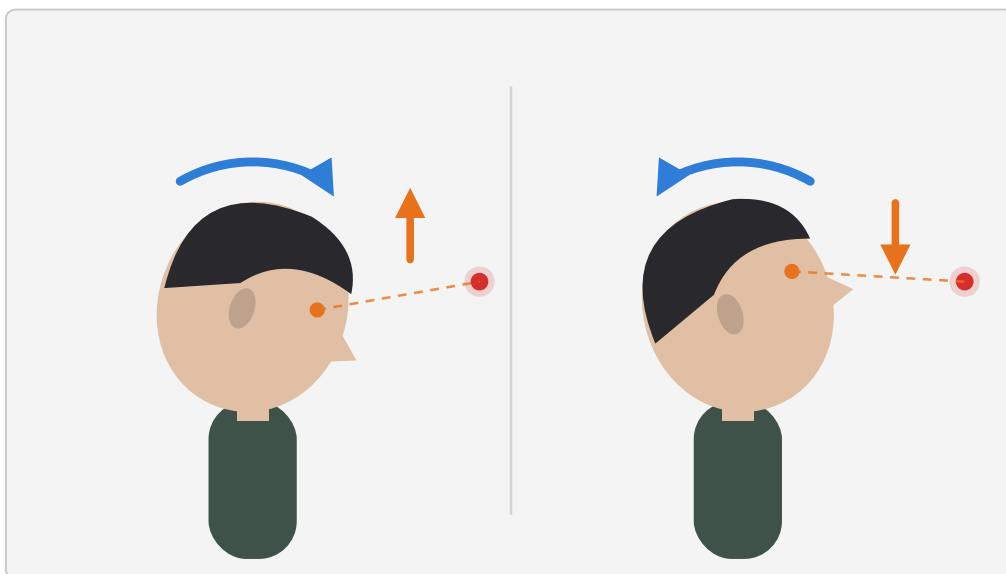
*The plane selector with LARP chosen: the drawn face turned 45° to the right and the guide below*

| Plane          | Head                                                | Impulse                | Canal                |
|----------------|-----------------------------------------------------|------------------------|----------------------|
| <b>Lateral</b> | facing forward, tilted slightly down                | sideways, right / left | right / left lateral |
| <b>LARP</b>    | turned 45° to the <b>right</b> , eyes on the target | nose down              | left anterior        |
|                |                                                     | nose up                | right posterior      |
| <b>RALP</b>    | turned 45° to the <b>left</b> , eyes on the target  | nose down              | right anterior       |
|                |                                                     | nose up                | left posterior       |



*With the head turned 45°, the LARP plane points at the target*

In the vertical planes the impulse is a **pitch** —the nose goes down or up— with the head already turned 45°. As always: short, abrupt, 10 to 20°, and a slow return. Hands on top of the head and under the chin, away from the phone.



*Side view: when the head goes down, the eye goes up to keep looking at the target, and vice versa*

If the impulse strays more than 30° from the plane —for example, pitching with the head facing forward in LARP— the impulse is rejected: **OFF THE PLANE**.

## 5. What to look at in the face

The iris has fibres and a dark crypt, and the sclera has vessels: with them you can see whether the eye **rolls** (torsion). With the head at 45° and the gaze in the plane of the canal, the compensating eye moves almost only up or down. With the head facing forward, the same turn would be half torsion: that is why the head is turned.

With a **Simulator** profile on, the affected canal shows the low gain and the saccades. Common vestibular neuritis affects the lateral and the anterior canal of one ear and spares the posterior.

## If something goes wrong

| You see                                             | What to do                                                                        |
|-----------------------------------------------------|-----------------------------------------------------------------------------------|
| Nose down lands in the “nose up” panel              | The screen faces the skin: turn the phone round and <b>Center</b> .               |
| The drawn face tilts sideways when the head pitches | The phone is on the side of the head: move it to the forehead and <b>Center</b> . |
| Many <b>REBOUND</b> or <b>OFF THE PLANE</b>         | The phone wobbles: adjust the mount or add non-slip mat.                          |
| The guide never turns green                         | <b>Center</b> with the head looking at the target, and only then turn it 45°.     |
| The computer says <b>RECONNECTING</b>               | The phone slept or lost wifi: it reconnects by itself when woken.                 |

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